

SASIDHAR MOPURU

Data & AI Platform Engineer | Databricks Certified

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PROFESSIONAL SUMMARY

Data Platform Engineer with 2.7 years of experience building real-time streaming and lakehouse systems using PySpark, Apache Kafka, Delta Lake, and Databricks. Proven track record of shipping modular, test-driven Python data pipelines with CI/CD. Recently built an end-to-end RAG Document QA application with FastAPI, Streamlit, ChromaDB, and LLMs, bridging data engineering into AI systems.

TECHNICAL SKILLS

- Languages: Python, SQL, PySpark
- Data Engineering: Apache Kafka, PySpark Streaming, ETL/ELT Pipelines, Data Ingestion, Data Transformation, Batch & Streaming Processing
- AI / LLM: RAG, LLM Integration, FastAPI, Streamlit, ChromaDB, Vector Databases, LangChain
- Big Data & Cloud: Databricks, Apache Spark, Delta Lake, Cloud Data Engineering Concepts
- Databases: SingleStore, Relational Databases
- Quality & Testing: Pytest, Unittest, Mocking, Data Validation, Schema Validation
- DevOps & Tools: Git, GitHub Actions, CI/CD Pipelines, IOMate, Jira, Agile Scrum
- Dev Practices: Object-Oriented Programming, Modular Coding, YAML, JSON, Logging, Exception Handling, System Design

PROFESSIONAL EXPERIENCE

Accenture | Data Engineer

Feb 2024 - Present

- Designed and developed scalable Source Data Product (SDP) and Core Data Product (CDP) pipelines for reliable data ingestion and transformation.
- Built reusable Python frameworks with configuration-driven architecture using YAML and JSON, reducing code duplication and improving maintainability.
- Developed Kafka producer and consumer applications for real-time data ingestion and event-driven processing.
- Implemented PySpark-based streaming solutions and gained practical exposure to Databricks data engineering workflows.
- Created comprehensive unit tests using Pytest, Unittest, and Mocking frameworks, improving application stability and reducing defects.
- Performed schema validation, data quality checks, and end-to-end testing to ensure accurate data processing.
- Managed version control and deployment workflows using Git and IOMate CI/CD pipelines.
- Collaborated with developers, testers, and business teams in Agile sprint cycles, participating in planning, development, testing, and release activities.

SELECTED PROJECTS

RAG Document QA Chatbot

github.com/Sasireddy001/rag-document-qa

- Built an end-to-end RAG application with FastAPI, Streamlit, ChromaDB, and OpenAI/Ollama LLMs for document question-answering.
- Implemented document ingestion, text chunking, sentence-transformer embeddings, dense retrieval, and LLM answer generation.
- Designed a modular, environment-driven architecture with separate ingest, embed, vector-store, LLM, and query components.
- Created a pytest test suite with mocked components and GitHub Actions CI.

Production-Style Kafka PySpark Delta Pipeline

github.com/Sasireddy001/Kafka-pyspark-delta-pipeline

- Built a production-style streaming data pipeline ingesting JSON events from Apache Kafka, transforming them with PySpark Structured Streaming, and writing results to Delta Lake.
- Implemented environment-driven configuration, JSON schema enforcement, checkpointing with exactly-once semantics, and watermark-based deduplication.
- Created a pytest unit-test suite with an in-memory Spark fixture, GitHub Actions CI, a sample data generator, and a throughput benchmark (~45k rows/sec on a 4-core laptop).
- Designed to run on Databricks, a Spark cluster, or locally for development.

CERTIFICATIONS

- Databricks Certified Data Engineer Associate - Databricks
- PySpark Streaming Training (Databricks) - Accenture
- Google Data Analytics Professional Certificate - Coursera
- NPTEL - Management Information Systems, IIT Madras

EDUCATION

- B.Tech - Computer Science & Technology | MITS, Madanapalle | JNTUA | CGPA: 8.0